Cluster and Cloud Computing – Lectures Week 4

Cloud Computing, OpenStack & Introducing MRC/NeCTAR

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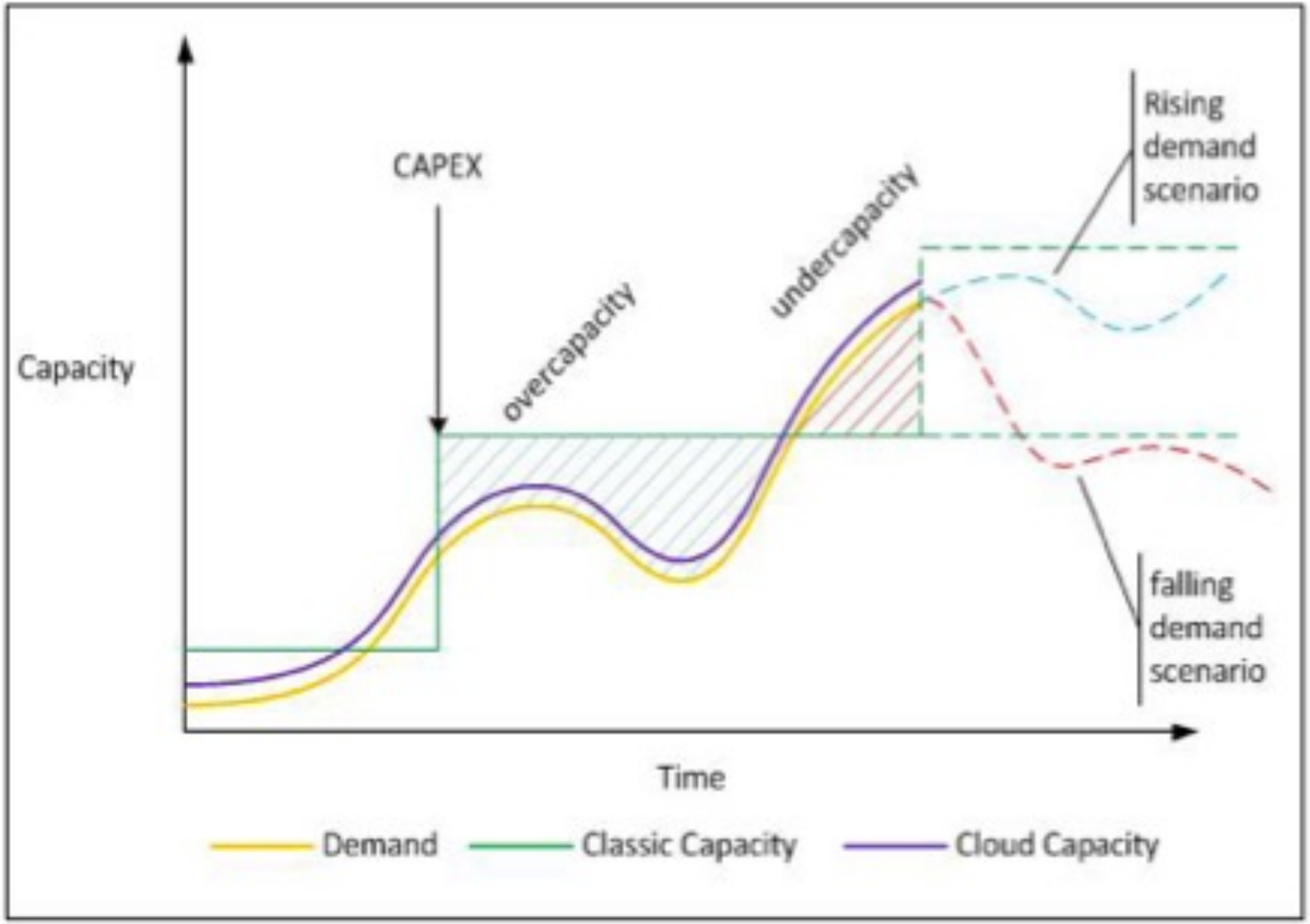
Overview

- Cloud benefits
  - Cloud marketing!?
- The various flavours of cloud
  - Introduction to #aaS?
- Break
  - Background to openStack
  - NeCTAR / MRC realization
    - Taster for workshops

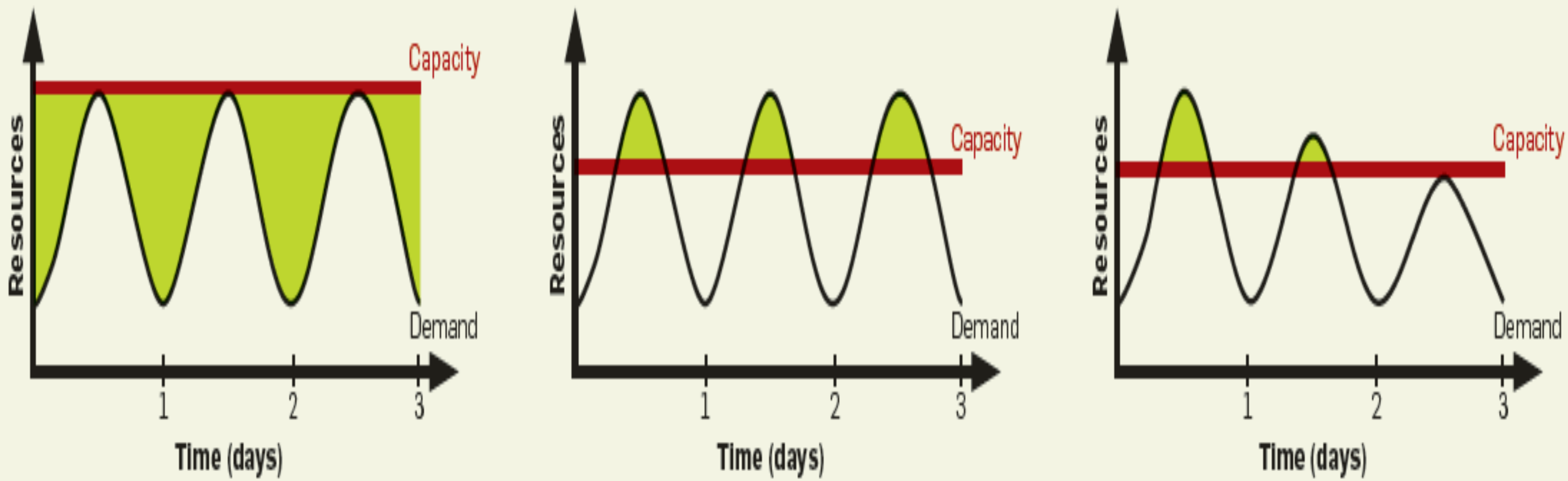
Life before cloud computing



Capacity vs Utilization curves [8]

Life before cloud computing…ctd

Figure 2. (a) Even if peak load can be correctly anticipated, without elasticity we waste resources (shaded area) during nonpeak times. (b) Underprovisioning case 1: potential revenue from users not served (shaded area) is sacrificed. (c) Underprovisioning case 2: some users desert the site permanently after experiencing poor service; this attrition and possible negative press result in a permanent loss of a portion of the revenue stream.



(a) Provisioning for peak load

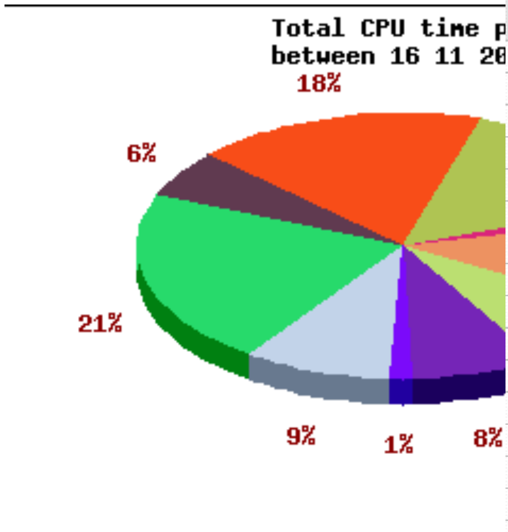
(b) Underprovisioning 1

(c) Underprovisioning 2

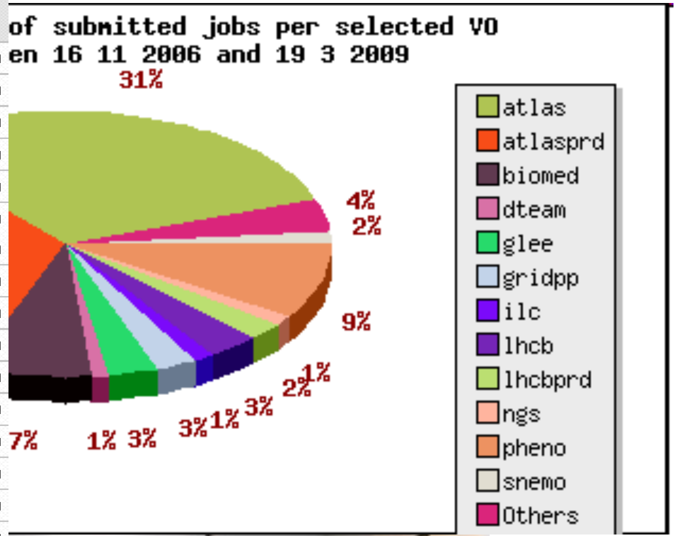
source: Ambrust et al, doi:10.1145/1721654.1721672

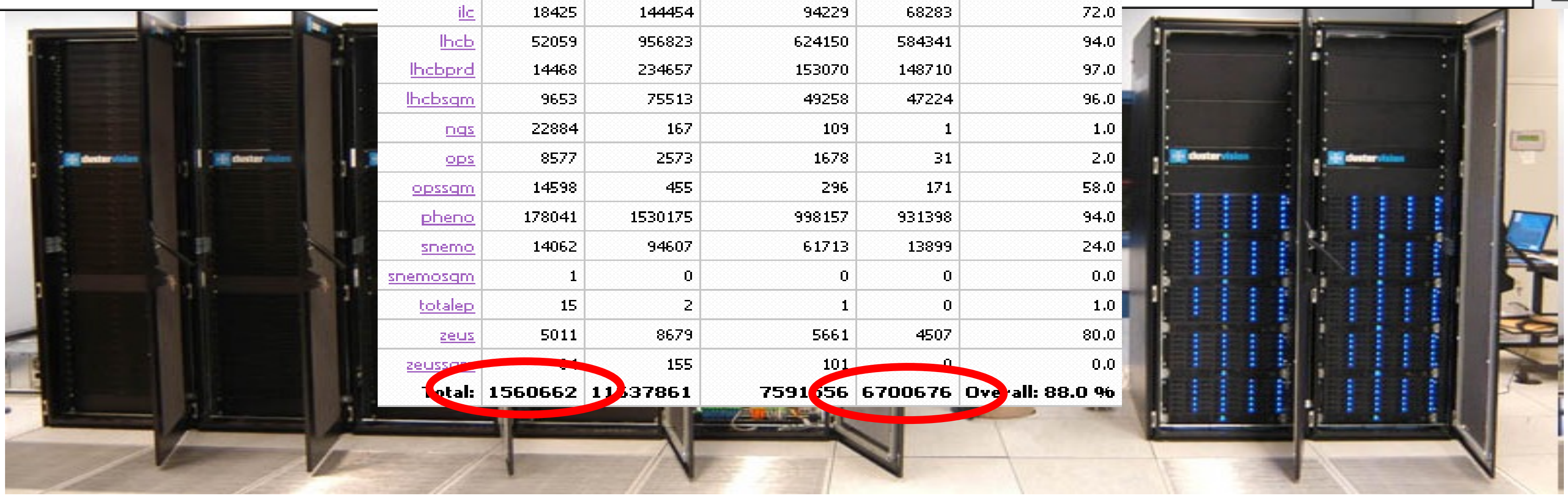
Cloud-busting? (~2009)

- To buy or not to buy that is the question…?
  - ScotGrid (www.scotgrid.ac.uk)
    - Cost £55



Group	Jobs	Kilo SPEC hours	WallClock Hours	CPU Hours	Eff (%)
alice	5	1	1	0	1.0
atlas	466733	2179463	1421698	1050451	74.0
atlasprd	497669	1808452	1179682	1091466	93.0
atlassgm	6420	930	606	465	77.0
babar	2	0	0	0	1.0
biomed	106413	1023678	667761	458138	69.0
biomedsgm	15	0	0	0	5.0
camont	9164	35510	23164	20070	87.0
cms	4857	18557	12105	6686	55.0
cmsprd	677	365	238	138	58.0
dteam	16494	3066	2000	344	17.0
dzero	92	141	92	75	82.0
glbio	33	694	453	446	99.0
glee	58306	2361972	1540752	1527659	99.0
gridpp	55924	1156772	754581	746173	99.0
ilc	18425	144454	94229	68283	72.0
lhcb	52059	956823	624150	584341	94.0
lhcbprd	14468	234657	153070	148710	97.0
lhcbsgm	9653	75513	49258	47224	96.0
ngs	22884	167	109	1	1.0
ops	8577	2573	1678	31	2.0
opssgm	14598	455	296	171	58.0
pheno	178041	1530175	998157	931398	94.0
snemo	14062	94607	61713	13899	24.0
snemosgm	1	0	0	0	0.0
totalep	15	2	1	0	1.0
zeus	5011	8679	5661	4507	80.0
zeussgm	[illegible]	155	101	0	0.0
Total:	1560662	11[illegible]37861	7591[illegible]56	6700676	Overall: 88.0 %





Cloud-busting? (~2009)

On-Demand Instances

United States | Europe

Standard On-Demand Instances	Linux/UNIX Usage	Windows Usage
1.7Gb memory/160GB disk	$0.11 per hour	$0.135 per hour
7.5Gb memory/850GB disk	$0.44 per hour	$0.54 per hour
15Gb memory/1690GB disk	$0.88 per hour	$1.08 per hour
High CPU On-Demand Instances	**Linux/UNIX Usage**	**Windows Usage**
Medium	$0.22 per hour	$0.32 per hour
Extra Large	$0.88 per hour	$1.28 per hour

Amazon Elastic Block Store

United States | Europe

Amazon EBS Volumes

- $0.11 per GB-month of provisioned storage
- $0.11 per 1 million I/O requests

Amazon EBS Snapshots to Amazon S3 (priced the same as Amazon S3)

- $0.18 per GB-month of data stored
- $0.012 per 1,000 PUT requests (when saving a snapshot)
- $0.012 per 10,000 GET requests (when loading a snapshot)

Data Transfer

Internet Data Transfer

The pricing below is based on data transferred "in" and "out" of Amazon EC2.

Data Transfer In	
All Data Transfer	$0.10 per GB

Data Transfer Out	
First 10 TB per Month	$0.17 per GB
Next 40 TB per Month	$0.13 per GB
Next 100TB per Month	$0.11 per GB
Over 150 TB per Month	$0.10 per GB

- $1=£0.69 (back then!)
  - £0.30*6,700,676 CPU hours
    - = £2,010,202 for just compute on-demand + data + networking + …?
- Now…???
  - https://aws.amazon.com/ec2/pricing/

Cloud Compute Cost Comparison 2025

On-Demand Pricing

Instance Type	AWS		Azure		Google		NeCTAR*
	Instances	Pricing (per hour)	VMs	Pricing (per hour)	VMs	Pricing (per hour)	
General purpose	m6g.xlarge	$0.1540	B4MS	$0.1660	e2-standard-4	$0.150924	$0
Compute optimized	c6g.xlarge	$0.1360	F4s v2	$0.1690	c2-standard-4	$0.2351	$0
Memory optimized	r6g.xlarge	$0.2016	E4a v4	$0.2520	m1-ultramem-40	$7.104	$0
Accelerated computing	p2.xlarge	For linux: $0.90 For Windows: $1.084	NC4as T4 v3	$0.5260	a2-highcpu-1g	$5.7077325	$0

https://www.flexera.com/blog/finops/aws-vs-azure-vs-google-cloud-pricing-compute-instances/

- Prices vary depending on
  - Supply/Demand
  - Location
- Complexity/Obfuscation
  - AWS - *Reserved Instances*
  - Azure - *Reserved Savings*
  - Google - *Commitment Price*
- Per hour/minute/second billing?
  - *Some locations*
  - *Some resources*
  - *Not others*

Cloud Storage Cost Comparison 2025

AWS S3 Storage

Amazon S3 | Overview | Features | Storage classes | Pricing | Security | Resources | FAQs

Region: US East (Ohio)

	Storage pricing
S3 Standard - General purpose storage for any type of data, typically used for frequently accessed data	
First 50 TB / Month	$0.023 per GB
Next 450 TB / Month	$0.022 per GB
Over 500 TB / Month	$0.021 per GB
S3 Intelligent - Tiering* - Automatic cost savings for data with unknown or changing access patterns	
Monitoring and Automation, All Storage / Month (Objects > 128 KB)	$0.0025 per 1,000 objects
Frequent Access Tier, First 50 TB / Month	$0.023 per GB
Frequent Access Tier, Next 450 TB / Month	$0.022 per GB
Frequent Access Tier, Over 500 TB / Month	$0.021 per GB
Infrequent Access Tier, All Storage / Month	$0.0125 per GB
Archive Instant Access Tier, All Storage / Month	$0.004 per GB
S3 Intelligent - Tiering* - Optional asynchronous Archive Access tiers	
Archive Access Tier, All Storage / Month	$0.0036 per GB
Deep Archive Access Tier, All Storage / Month	$0.00099 per GB
S3 Standard - Infrequent Access** - For long lived but infrequently accessed data that needs millisecond access	
All Storage / Month	$0.0125 per GB
S3 One Zone - Infrequent Access** - For re-createable infrequently accessed data that needs millisecond access	
All Storage / Month	$0.01 per GB
S3 Glacier Instant Retrieval*** - For long-lived archive data accessed once a quarter with instant retrieval in milliseconds	
All Storage / Month	$0.004 per GB
S3 Glacier Flexible Retrieval (Formerly S3 Glacier)***- For long-term backups and archives with retrieval option from 1 minute to 12 hours	
All Storage / Month	$0.0036 per GB
S3 Glacier Deep Archive*** - For long-term data archiving that is accessed once or twice in a year and can be restored within 12 hours	
All Storage / Month	$0.00099 per GB

NeCTAR*

$0

Azure Blob Storage

File Structure: Hierarchical Namespace (NFS v3.0, SFTP Pr
Redundancy: LRS
Region: West US 2
Currency: United States – Dollar ($) USD

Data storage prices pay-as-you-go

All prices are per GB per month.

Data storage prices pay-as-you-go	Premium	Hot	Cool	Archive
First 50 terabyte (TB) / month	$0.15 per GB	$0.018 per GB	$0.01 per GB	$0.00099 per GB
Next 450 TB / month	$0.15 per GB	$0.0173 per GB	$0.01 per GB	$0.00099 per GB
Over 500 TB / month	$0.15 per GB	$0.0166 per GB	$0.01 per GB	$0.00099 per GB

Google Storage

Click on a geographic area to view the at-rest costs for associated locations:

Regions | Dual-regions | Multi-regions

North America | South America | Europe | Middle East | Asia | Indonesia | Australia

Location	Standard storage (per GB per Month)	Nearline storage (per GB per Month)	Coldline storage (per GB per Month)	Archive storage (per GB per Month)
Iowa (us-central1)	$0.020	$0.010	$0.004	$0.0012
South Carolina (us-east1)	$0.020	$0.010	$0.004	$0.0012
Northern Virginia (us-east4)	$0.023	$0.013	$0.006	$0.0025
Columbus (us-east5)	$0.020	$0.010	$0.004	$0.0012
Oregon (us-west1)	$0.020	$0.010	$0.004	$0.0012
Los Angeles (us-west2)	$0.023	$0.016	$0.007	$0.0025
Salt Lake City (us-west3)	$0.023	$0.016	$0.007	$0.0025
Las Vegas (us-west4)	$0.023	$0.013	$0.006	$0.0025
Dallas (us-south1)	$0.020	$0.010	$0.004	$0.0012
Montréal (northamerica-northeast1)	$0.023	$0.013	$0.007	$0.0025
Toronto (northamerica-northeast2)	$0.023	$0.013	$0.007	$0.0025

https://www.cloudzero.com/blog/cloud-storage-pricing/

Other Cloud Costs

- Networking Costs
  - Ingress vs egress fees
    - Egress typically far more costly than input
  - Average of 6% of Cloud costs
    - https://www.computerweekly.com/feature/Cloud-egress-costs-What-they-are-and-how-to-dodge-them
- Services
  - Not just compute/storage
  - Licensed software
- Challenging to know all of this in advance

Cloud Computing: A Definition

- NIST definition: "Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) <u>that can be rapidly provisioned and released with minimal management effort or service provider interaction</u>."
  - » National Institute of Standards and Technology (http://dx.doi.org/10.6028/NIST.SP.800-145)

The Most Common Cloud Models

Deployment Models

- Private
- Community
- Public
- Hybrid

Delivery Models

- Software as a Service (SaaS)
- Platform as a Service (PaaS)
- Infrastructure as a Service (IaaS)

Essential Characteristics

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

Public Clouds

- Pros
  - Utility computing
  - Can focus on core business
  - Cost-effective
  - "Right-sizing"
  - Democratisation of computing
- Cons
  - Security
  - Loss of control
  - Possible lock-in
  - Dependency of Cloud provider continued existence

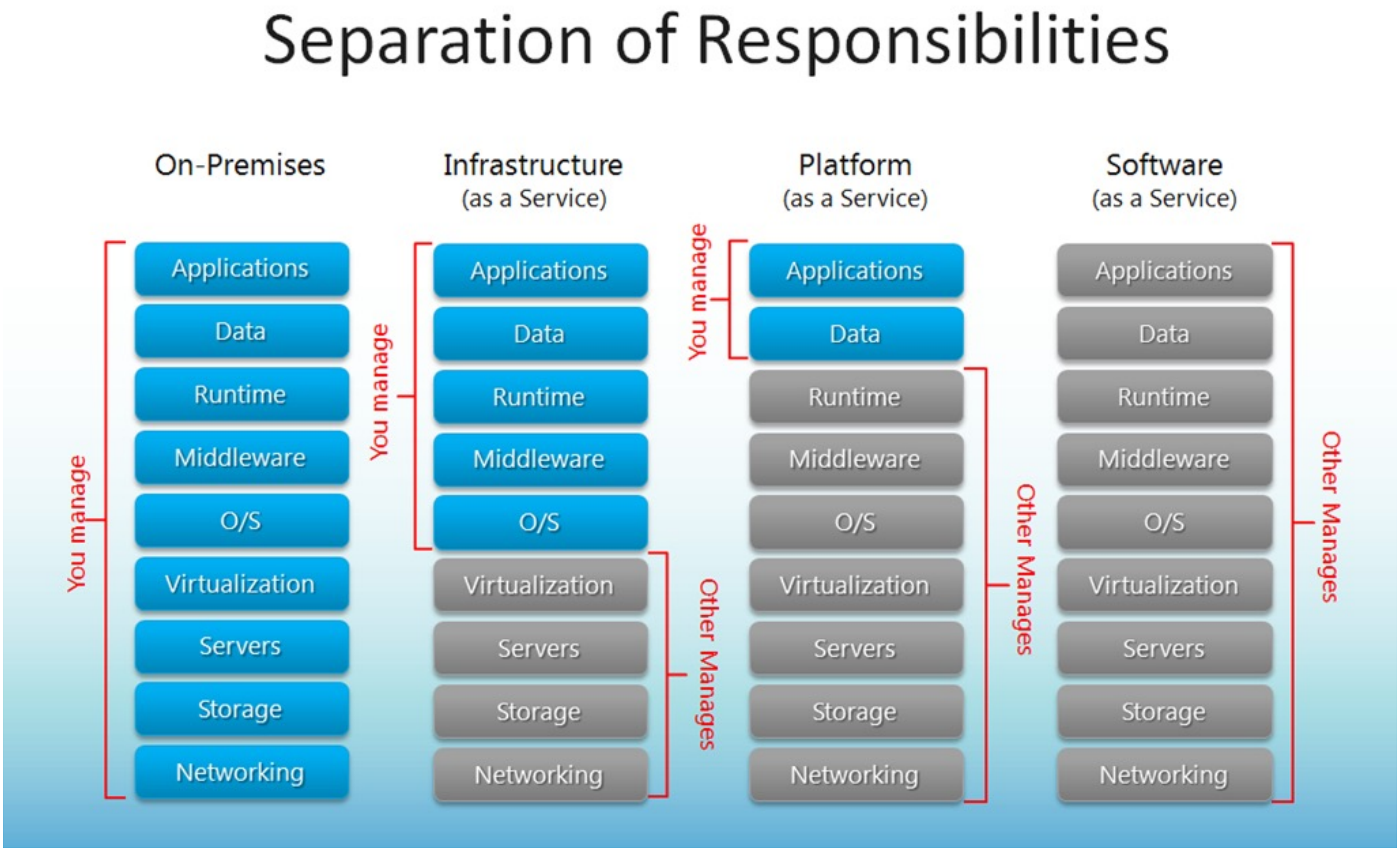
Private (on premise) Clouds

- Pros
  - Control
  - Consolidation of resources
  - Easier to secure
  - More trust
- Cons
  - Relevance to core business?
    - e.g., Netflix ->Amazon
  - Staff/management overheads
  - Hardware obsolescence
  - Over/under utilisation challenges
  - (Effort and cost of establishing a data centre)

Hybrid Clouds

- Pros
  - Cloud-bursting
    - Use private cloud, but burst into public cloud when needed
- Cons
    - How do you move data/resources when needed?
    - How to decide (in real time?) what data can go to public cloud?
    - Short term need can be much more expensive
    - Is the public cloud compliant with PCI-DSS (Payment Card Industry – Data Security Standard)?
- Examples
  - Eucalyptus, VMWare Cloud Foundation (vSphere)

*aaS Delivery Models



Public SaaS examples

- Gmail
- Sharepoint
- Salesforce.com CRM
- On-live
- Gaikai
- Microsoft Office 365
- Many, many other examples

Public PaaS Examples

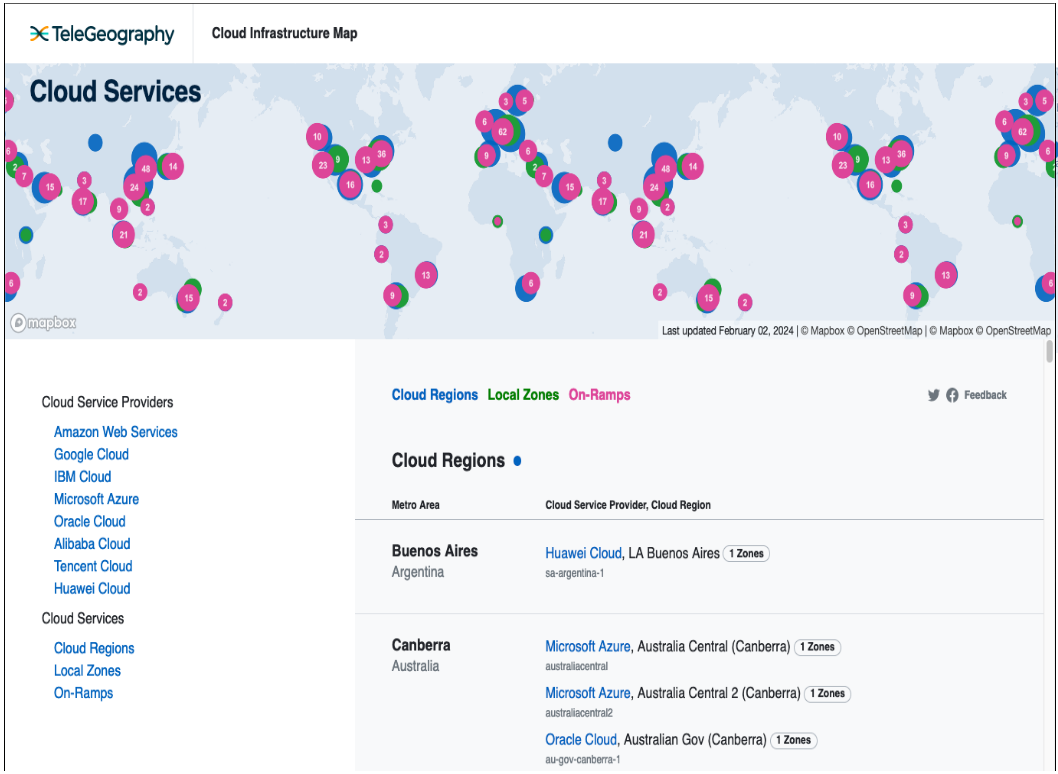
Cloud Name	Language and Developer Tools	Programming Models Supported by Provider	Target Applications and Storage Options
Google App Engine	Python, Java, Go, PHP + JVM languages (scala, groovy, jruby)	MapReduce, Web, DataStore, Storage and other APIs	Web applications and BigTable storage
Salesforce.com's Force.com	Apex, Eclipsed-based IDE, web-based wizard	Workflow, excel-like formula, web programming	Business applications such as CRM
Microsoft Azure	.NET, Visual Studio, Azure tools	Unrestricted model	Enterprise and web apps
Amazon Elastic MapReduce	Hive, Pig, Java, Ruby etc.	MapReduce	Data processing and e-commerce
Aneka	.NET, stand-alone SDK	Threads, task, MapReduce	.NET enterprise applications, HPC

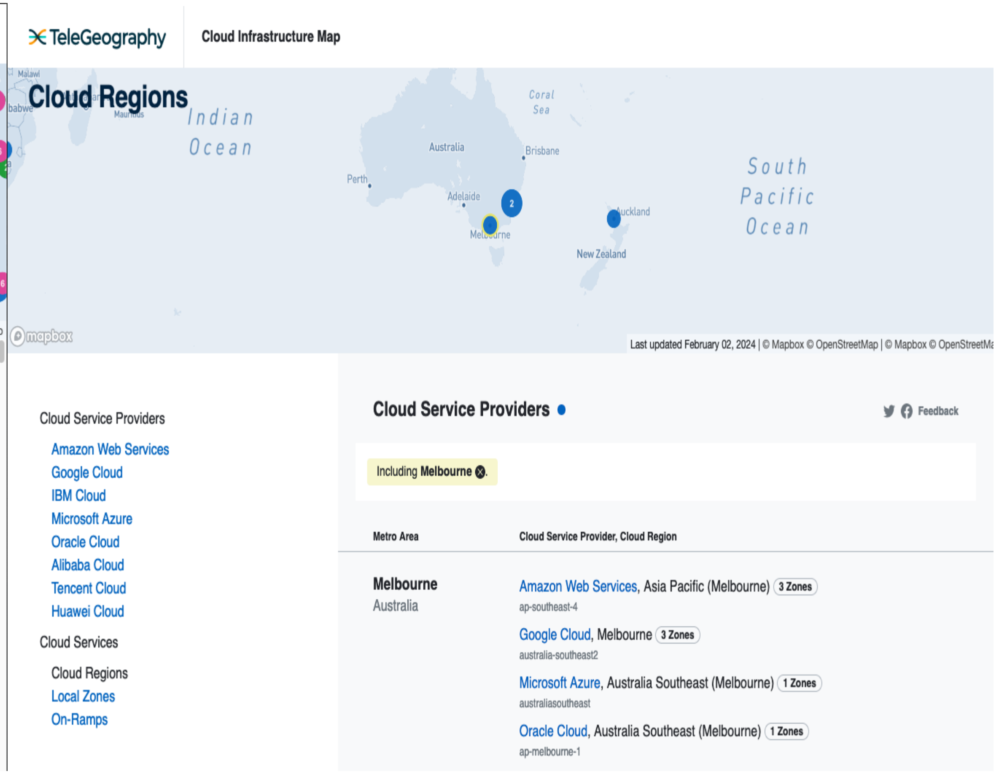
Infrastructure As A Service (IaaS)

- Primary focus of this course…
- Many providers
  - Amazon Web Services (Market leader)
    - http://aws.amazon.com
  - Azure
    - https://azure.microsoft.com/
  - Google
    - https://cloud.google.com/
  - *AliBaba, Catalyst, DigitalOcean, Dimension Data, GoGrid, Helio, Huawei, IBM, Joyent, Oracle, ProfitBricks, Rackspace, Tencent, Verizon, … and many, many others!*

Public Cloud Service Provider Mappings

- See https://www.cloudinfrastructuremap.com/





  - Melbourne Research Cloud
    - https://dashboard.cloud.unimelb.edu.au
  - NeCTAR Research Cloud
    - https://dashboard.rc.nectar.org.au

THE UNIVERSITY OF MELBOURNE

NeCTAR Research Cloud

nectar

- National eResearch Collaboration Tools and Resources (NeCTAR – www.nectar.org.au)
  - Established ~2010
  - $100m+ federal funding (Aus .gov)
  - Had four key strands
    - ~~National Servers Program~~
    - Research Cloud Program
      - OpenStack IaaS
      - 4Gb-64Gb (mostly Linux flavours)
      - 30,000 physical servers available across different availability zones
        - » Upgraded continually!
    - ~~eResearch Tools Program~~
    - Virtual Laboratories Program
      - Astro,
      - Genomics,
      - Humanities,
      - Climate,
      - Nano-,
      - …endocrine genomics

ARDC
Australian Research Data Commons

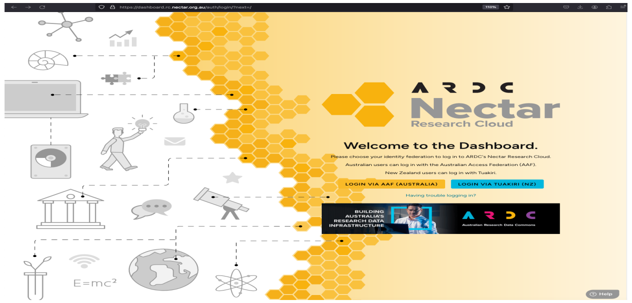
National Data Infrastructure

- Research Data Services (RDS)
  - $100m+ federal funding
  - Establish data storage resources across Australia
    - ~100 Petabytes national storage
  - Victoria Node (VicNode)
    - UniMelb, UniMonash for Vic-wide "nationally significant data sets"
      - Used by many diverse communities

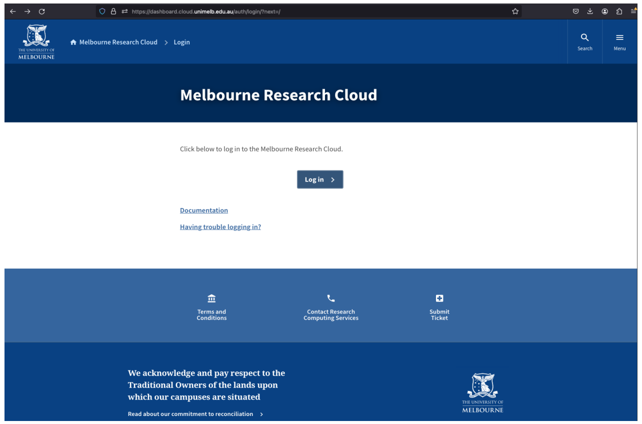
Australian Research Data Commons

Melbourne Research Cloud and NeCTAR

- NeCTAR (https://dashboard.rc.nectar.org.au)
  - National Cloud
  - Multiple availability zones
  - National projects, e.g. ARC, NHMRC
  - GPUs (one month at a time!)



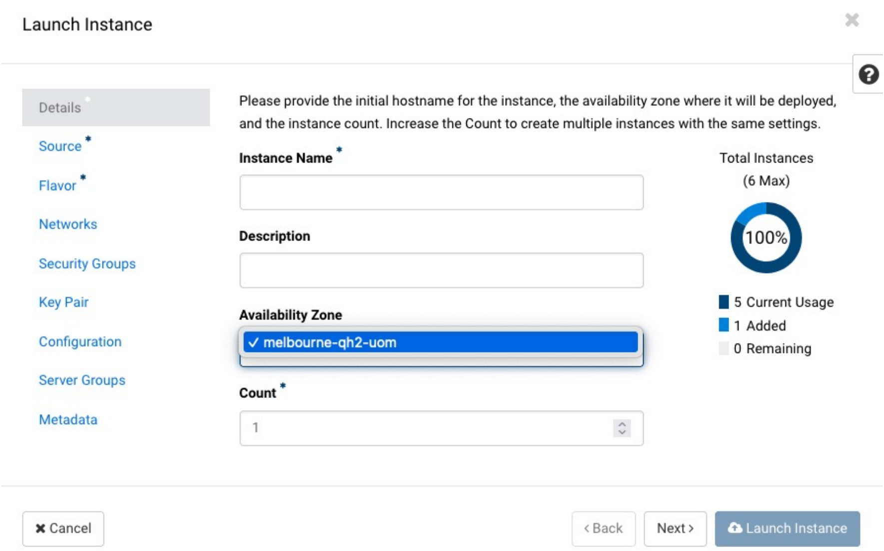
- MRC (https://dashboard.cloud.unimelb.edu.au)
  - UniMelb-only Cloud
  - Single data centre
  - Limited number of IP addresses
  - No GPUs
    - Go to SPARTAN



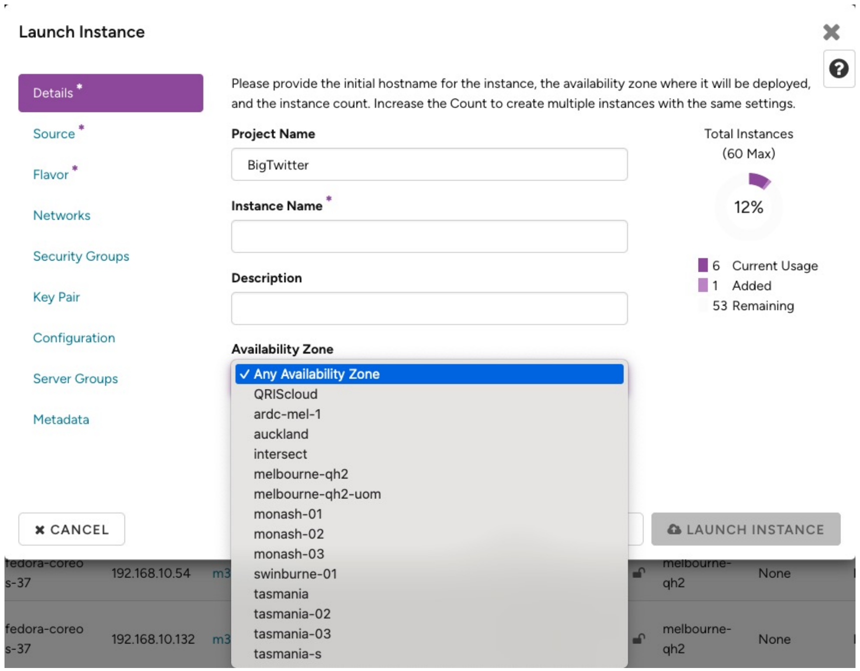
- Both accessible through the AAF
  - Both based on openStack – but not exact same deployments!!!

Availability Zones!?!

- MRC

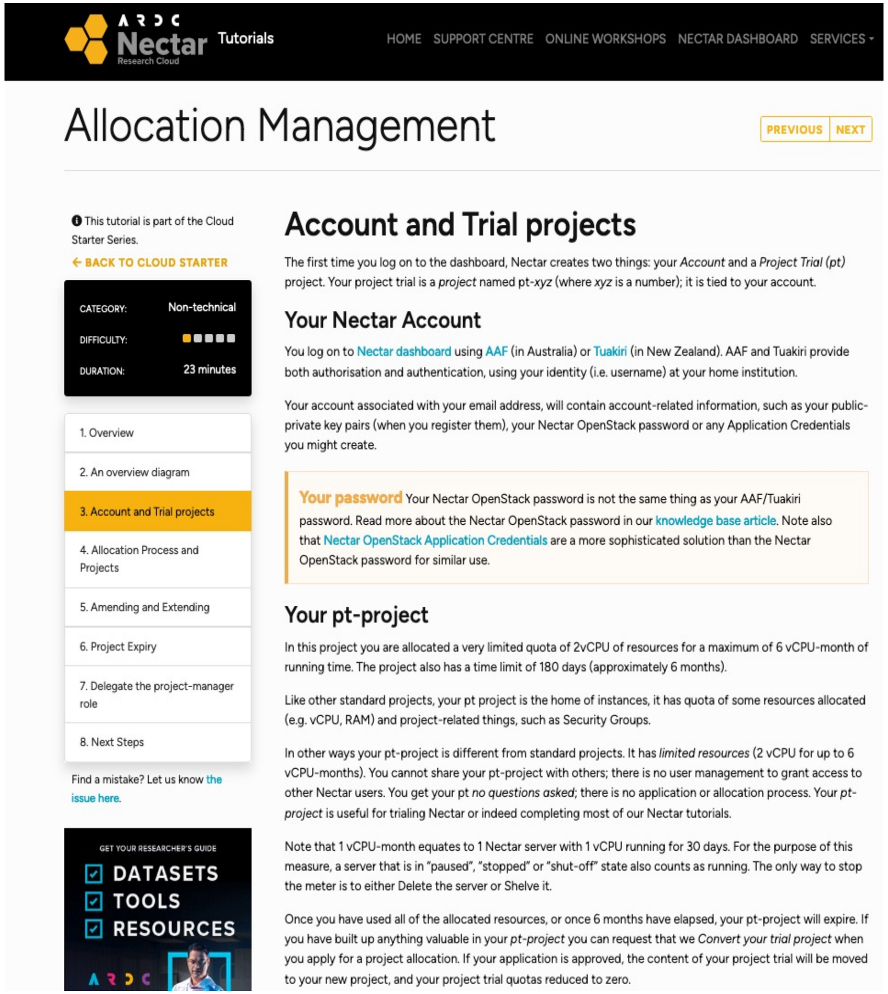


- NeCTAR



Free lunch on NeCTAR!?

- You can access and use it (if you haven't already)
  - https://tutorials.rc.nectar.org.au/allocation-management/03-account-and-trial
    - No allocation request needed
    - 2 vCPUs
      - 6 months total *usage* (2*3 months)
      - 180 days then stopped
        - » Unless extension request ok'd
      - No project sharing/user mgt
      - No volume storage
    - Free resource may be useful for ass2



BREAK

OpenStack

- Began in 2010 as a joint project between Rackspace and NASA
- Offers free and open-source software platform for cloud computing for (mostly) IaaS
- Consists of interrelated components (services) that control / support compute, storage, and networking resources
- Often used through web-based dashboards, through command-line tools, or programmatically through ReSTful APIs
- Released under Apache License
- Managed/coordinated by the OpenStack Foundation
  - non-profit corporate entity established in 2012 to promote OpenStack software and its community
  - Over 500 companies have since joined the project

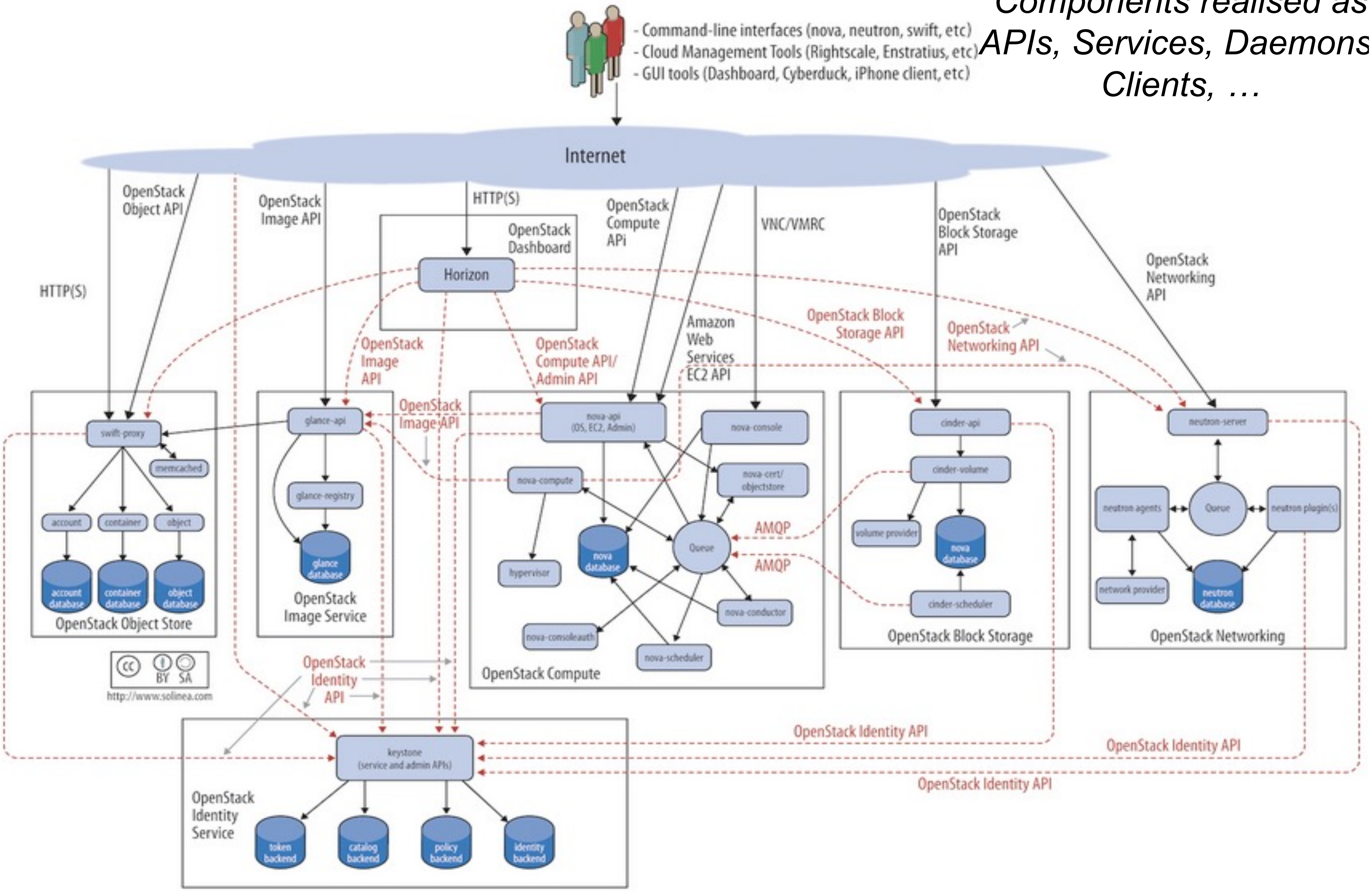
OpenStack Components

- Many associated/underpinning services
  - Security Management (code-named Keystone)
  - Dashboard service (code-named Horizon)
  - Compute Service (code-named Nova)
  - Network Service (code-named Neutron)
  - Image Service (code-named Glance)
  - Block Storage Service (code named Cinder)
  - Object Storage Service (code-named Swift)
  - *Orchestration Service (code-named Heat)*
  - *Container Service (code-named Zun)*
  - *Database service (code-named Trove)*
  - …

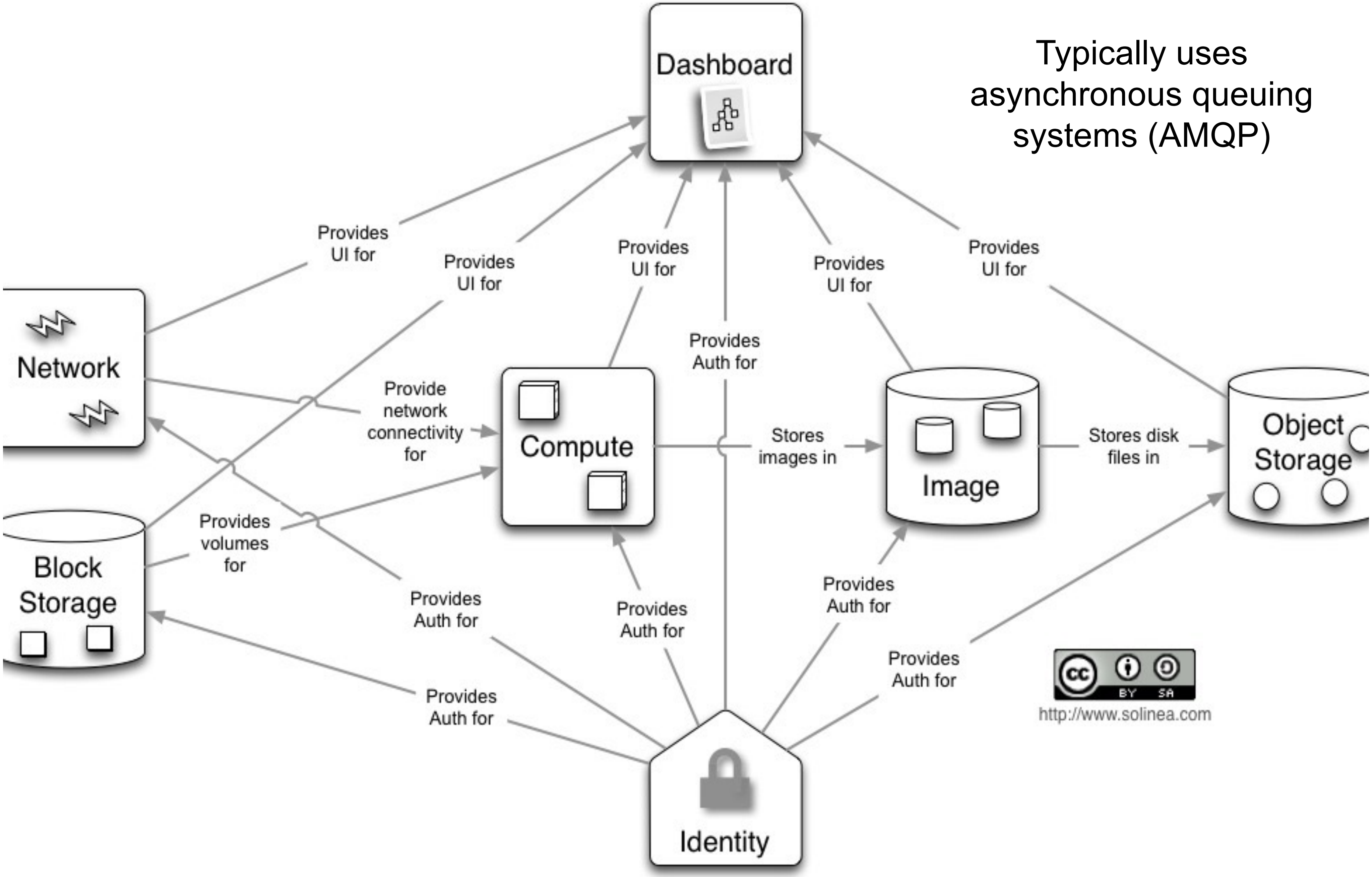
https://www.openstack.org/software/project-navigator/openstack-components#openstack-services

(Simplified) OpenStack Architecture

Components realised as APIs, Services, Daemons Clients, …



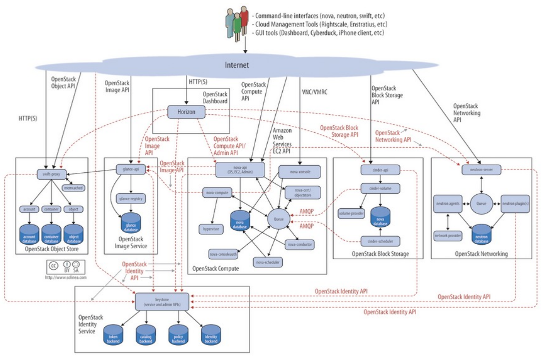
(Simplified) User Perspective



Typically uses asynchronous queuing systems (AMQP)

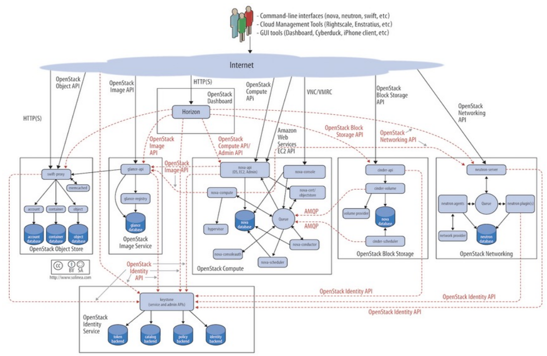
Key Services::Identity Service

- Keystone
  - Provides an authentication and authorization* service for OpenStack services
    - Tracks users/permissions
  - Provides a catalog of endpoints for all OpenStack services
    - Each service registered during install
      - Know where they are and who can do what with them
    - Project membership; firewall rules; image mgt; …
  - *Generic authorization system for openStack…
    - more in security lecture
  - NOTE
    - Email Aliases and Assignment 2!



Key Services::Dashboard

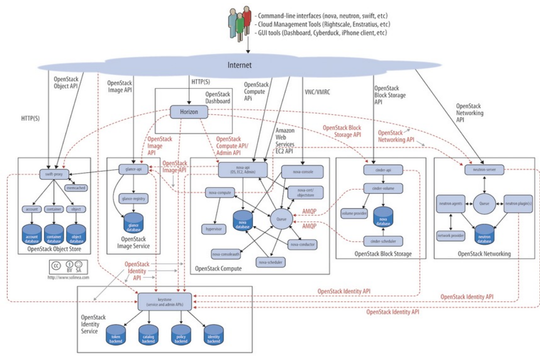
- Horizon
  - Provides a web-based self-service portal to interact with underlying OpenStack services, such as launching an instance, assigning IP addresses and configuring access controls.
  - Based on Python/Django web application
  - Mod_wsgi
    - Apache plug realising web service gateway interface
  - Requires Nova, Keystone, Glance, Neutron
  - Other services optional…



Key Services::Compute

- Nova
  - Manages the lifecycle of compute instances in an OpenStack environment
  - Responsibilities include spawning, scheduling and decommissioning of virtual machines on demand
  - Virtualisation agnostic
    - Libvirt
      - Open-source API, daemon and tools for managing platform virtualisation including support for Kernel based virtual machine (KVM), Quick Emulator (QEMU), Xen, Lightweight Linux Container System (LXC)
    - XenAPI, Hyper-V, VMWare ESX,…
    - Docker
    - …

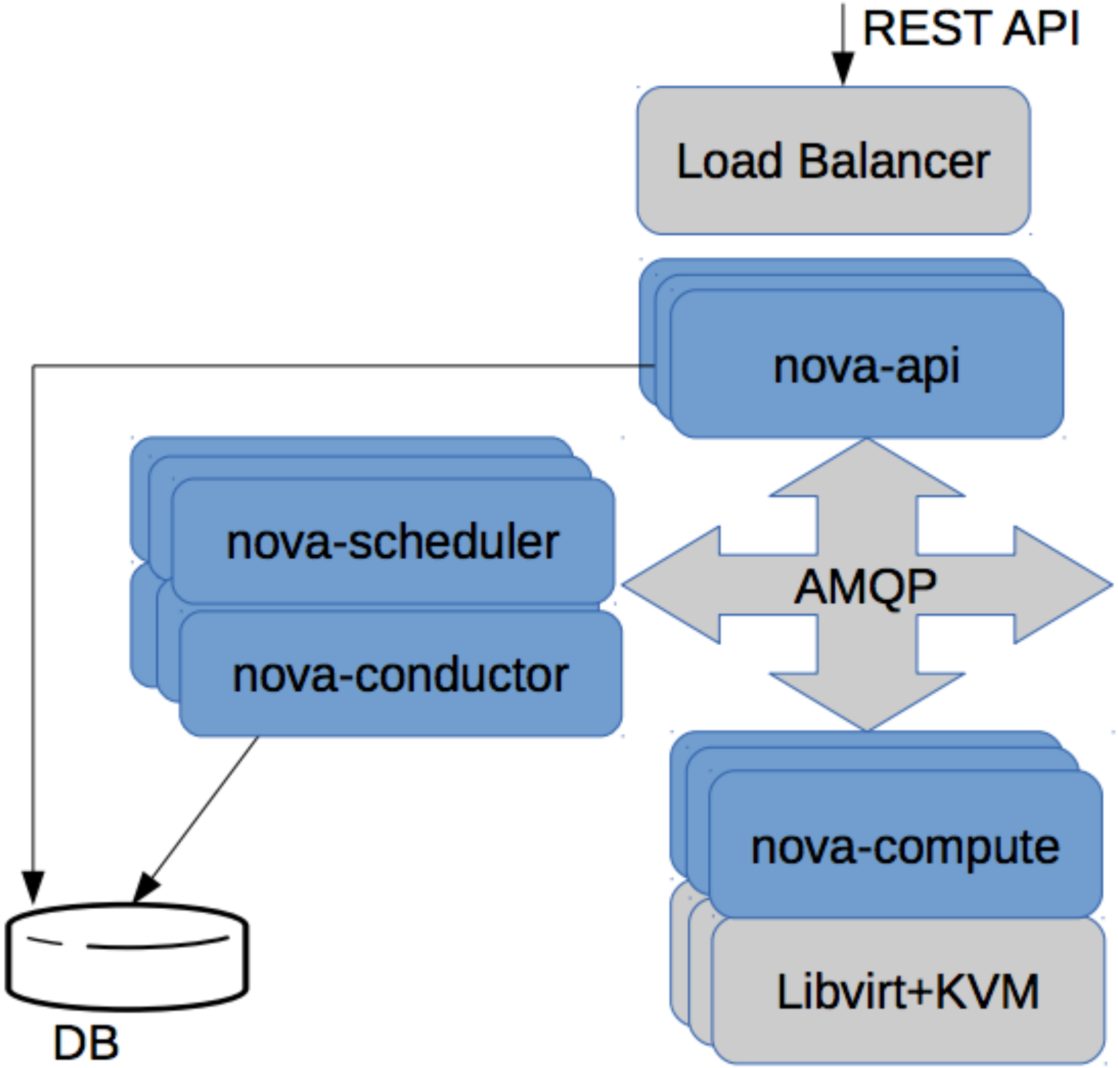
Virtualisation limits on MRC/NeCTAR?



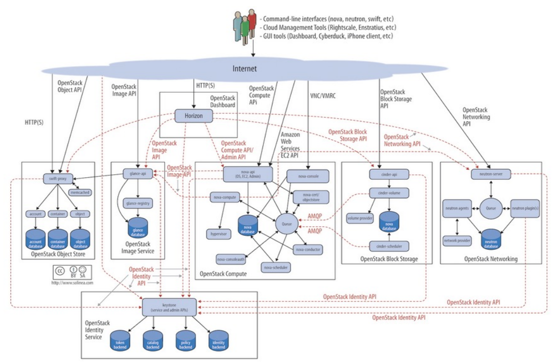
Key Services::Compute

- Nova
  - API
    - Nova-api - accepts/responds to end user API calls; supports openStack Compute & EC2 & admin APIs
  - Compute Core
    - Nova-compute - Daemon that creates/terminates VMs through hypervisor APIs
    - Nova-scheduler - schedules VM instance requests from queue and determines which server host to run
    - Nova-conductor - Mediates interactions between compute services and other components, e.g. image database
  - Networking
    - Nova-network - Accepts network tasks from queue and manipulates network, e.g. changing IPtable rules
  - Image Mgt, Client Tools, …

Simplified Nova Compute Scenario



I need a VM with:
- 64Gb memory,
- 8vCPUs,
- in Melbourne,
- running Ubuntu v23.10
- …

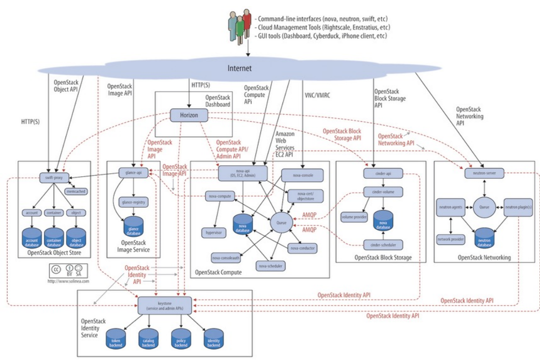


A Word about Cloud Storage

- Object storage
  - Suitable for large amounts of unstructured data (Pb+)
  - Data decomposed into discrete units (objects)
  - Can be text, images, values… and associated metadata
  - Metadata used for object access
    - unique (hashed) Id, name, size, creation data, other tags
    - flat namespace – direct access via Id
- Block storage
  - Suitable for high-speed processing (low latency)
  - Data into fixed sized individual blocks (few kB-Mb)
  - Each block has a unique address (logical block addressing)
  - Data lookup table for data access
- File storage
  - Suitable for concurrent access to shared files
    - Hierarchical storage system (human oriented)
    - Fine-grained access control
    - Common file access protocols

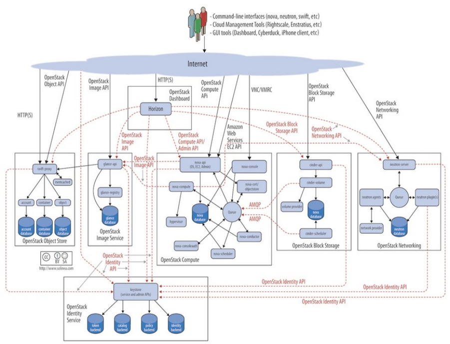
Key Services::Object Storage

- Swift
  - Stores and retrieves arbitrary unstructured data objects via RESTful API, e.g. VM images and data
    - Not POSIX (atomic operations); eventual consistency
  - Fault tolerant with data replication and scale-out architecture.
    - Available from anywhere; persists until deleted
    - Allows to write objects and files to multiple drives, ensuring the data is replicated across a server cluster
  - Can be used with/without Nova/compute
  - Client; admin support
    - e.g. Swift client – allows users to submit commands to ReST API through command line clients to configure/ connect object storage to VMs



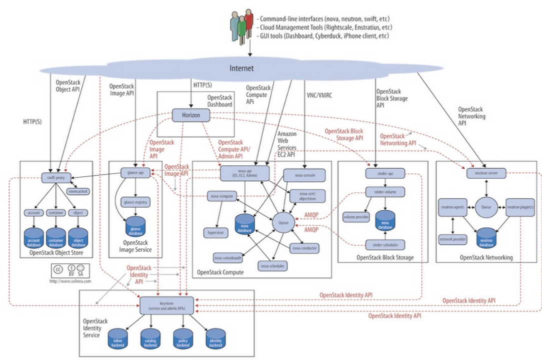
Key Services::Block Storage

- Cinder
  - Provides persistent block storage to virtual machines (instances) and supports creation and management of block storage devices
  - Cinder access associated with a VM
    - Cinder-api – routes requests to cinder-volume
    - Cinder-volume – interacts with block storage service and scheduler to read/write requests; can interact with multiple flavours of storage (flexible driver architecture)
    - Cinder-scheduler – selects optimal storage provider node to create volumes (ala nova-scheduler)
    - Cinder-backup – provides backup to any types of volume to backup storage provider
      - Can interact with variety of storage solutions



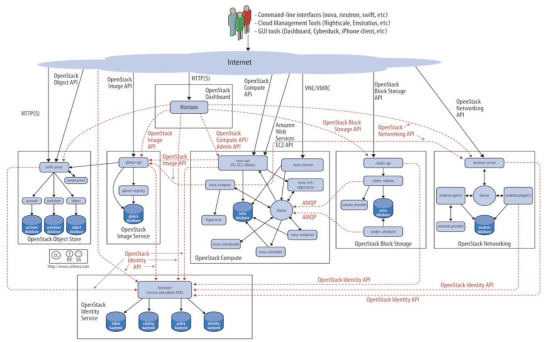
Key Services::Image Service

- Glance
  - Accepts requests for disk or server images and their associated metadata (from Swift) and retrieves / installs (through Nova)
    - Glance-api – image discovery, retrieval and storage requests
    - Glance-registry – stores, processes and retrieves metadata about images, e.g. size and type
      - Fedora-CoreOS-38?
      - NeCTAR Windows Server 2022?
      - NeCTAR R-Studio?
      - NeCTAR JupyterLab?
      - My last good snapshot…?
        - » (more in Workshops)



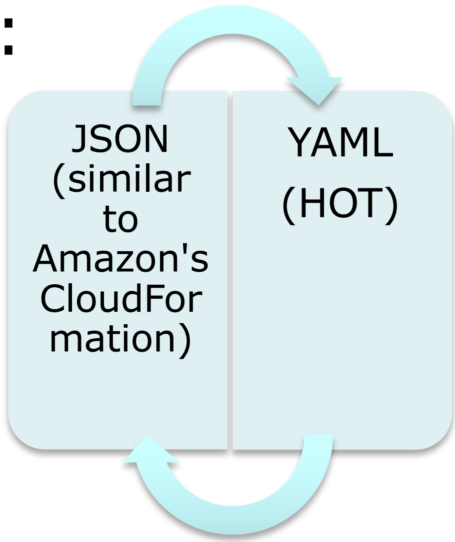
Key Services::Networking

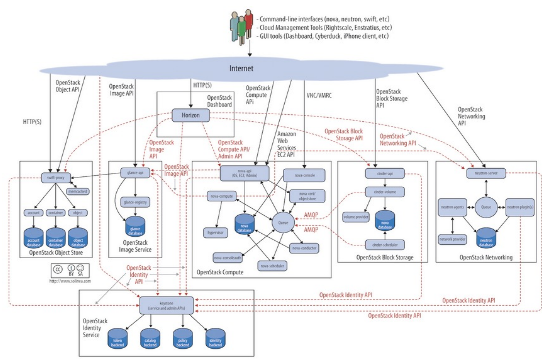
- Neutron
  - Supports networking of OpenStack services
  - Offers an API for users to define networks and the attachments into them, e.g. switches, routers
  - Pluggable architecture that supports multiple networking vendors and technologies
  - Neutron-server – accepts and routes API requests to appropriate plug-ins for action
    - Port management, e.g. default SSH, VM-specific rules, …
    - More broadly configuration of availability zone networking, e.g. subnets, DHCP, …



Key Services::Orchestration Service

- Heat
  - Template-driven service to manage lifecycle of applications deployed on Openstack
  - *Stack*: Another name for the template and procedure behind creating infrastructure and the required resources from the template file
  - Can be integrated with automation tools such as Chef, Puppet, Ansible, etc.
  - Template format:





To Do List

- Log in to MRC
  - https://dashboard.cloud.unimelb.edu.au/
- Select UniMelb from drop down list
  - Log in as yourself (University username/password)
    - Not password you selected for SPARTAN!
  - Once done you will turn orange in due course in Googlesheet which means we can add you (automatically to your teams)

Team 1	Team 2	Team 3	Team 4	Team 5
weizhi.quan@student.unimelb.edu.au	rzhou4373@student.unimelb.edu.au	qingbot@student.unimelb.edu.au	jake.gerstel@student.unimelb.edu.au	fuwf1@student.unimelb.edu.au
zhosheng@student.unimelb.edu.au	xuej5@student.unimelb.edu.au	kaichenglian@student.unimelb.edu.au	zhangrh@student.unimelb.edu.au	shmeng@student.unimelb.edu.au
haxu4@student.unimelb.edu.au	renyuanf@student.unimelb.edu.au	xingcanw@student.unimelb.edu.au	wangyangw@student.unimelb.edu.au	yujin2@student.unimelb.edu.au
jqqian1@student.unimelb.edu.au	chaogez@student.unimelb.edu.au	kbalamurali@student.unimelb.edu.au	linyingwei@student.unimelb.edu.au	yanrr@student.unimelb.edu.au
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Introduction to Cloud

- Demonstration of the basic functionality of the Horizon dashboard on NeCTAR

- More in workshops later/Thursday